

Last One Standing

How much of Canada's drug supply thinned to a single company, counted from the regulator's own records, and the arithmetic that took our best finding away.

THE COUNT

molecules currently marketed	2,044
with exactly one supplier	1,203 (58.9 %)
of those, once had two or more	407
single-supplier with unexpired listed patent	343
ever on the Patent Register at all	513
Rx-only, no unexpired patent, pre-2006, one supplier	382
once marketed in Canada, no longer	6,651

407 is the project. Not how concentrated the market is today, but how much of it thinned while nobody was counting. Reconstructed from every product Health Canada has ever authorised, not inferred from a snapshot.

WE DO NOT HEADLINE 58.9 %

It is not ours and not new. PMPRB / Gaudette et al. (*JAMA Health Forum* 2024) report **52 %** of off-patent drugs in a monopoly market in 2022. Zhang et al. (*CMAJ Open* 2020) reach ≈ 51 %. We differ because we count active-ingredient sets, and because company_name is the authorisation holder, not the plant. Merging parents could only *lower* our number.

ONE SUPPLIER IS NOT A SHORTAGE WARNING

PMPRB's 2022 shortage study: of 8,558 shortage reports only **2 %** were single-source non-patented drugs, and single-source drugs were in shortage **22 %** of the time against **32 %** for multi-source. Santhireswaran et al. (*JAMA Netw Open* 2026) found drugs with more than 25 products available had the *higher* shortage intensity. If this page implied a supply warning, the regulator would contradict it. It does not.

FINISHED VS PLANNED

Finished: full pipeline from a clean clone; 25 checks at a fixed denominator; 25 planted defects, 25 caught, 0 missed; every figure here re-derived by a check, none typed by hand.
Not done: no shortage data (healthproductshortages.ca needs an account we did not create); rx reads today's value because the database has no point-in-time status; all cohorts share one endpoint and overlap heavily, so this is three views of one outcome, not three replications; no therapeutic-substitution check, so we cannot separate obsolescence from loss.

THE FINDING WE DELETED

Molecules with more suppliers disappear less often. That gradient holds in all three cohorts and it looks like a finding. **It is arithmetic.** A molecule leaves the market only when every supplier leaves, so if companies depart independently at rate p , the rate falls as p^k with no signal at all. Measuring p from the 2016 cohort gives 0.486:

suppliers 2016	n	gone	null p^k	diff
1	1,233	37.8 %	48.6 %	-10.8
2	279	27.2 %	23.6 %	+3.7
3	175	11.4 %	11.4 %	-0.0
4	106	5.7 %	5.6 %	+0.1
5	84	9.5 %	2.7 %	+6.8
6+	295	3.4 %	1.3 %	+2.1

8 cells above the null, 6 below, not one direction. The single-supplier cell is consistently *better* than the null: the opposite of the story the raw gradient tells. src/null.py is the script that took it from us, and it ships in the repo.

DOES ANYTHING BEAT COUNTING?

cohort	n	gone	suppliers only	model	diff
2011	2,120	748	0.690	0.754	+0.064
2016	2,172	586	0.681	0.758	+0.078

Logistic regression in pure Python, no numpy, fit on 2006 and tested on later cohorts. Features restricted to what was knowable in the cohort year. **The strongest positive weight is lost_already (+0.813):** companies a molecule has *already* shed, above holders (-0.908) and products_by_then (-0.670). Thinning predicts disappearing better than thinness does.

Health Canada Drug Product Database (bulk JSON, no key) + Patent Register. Python 3 standard library only: no packages, no API keys, no accounts. The page opens from the filesystem. Retrieval dates recorded in src/sources.py.